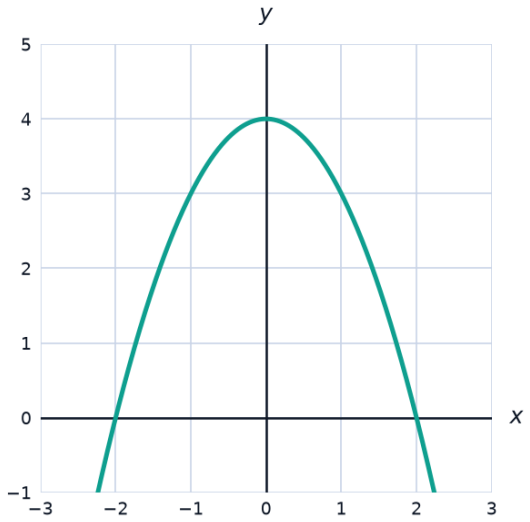
Area Between Curves

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

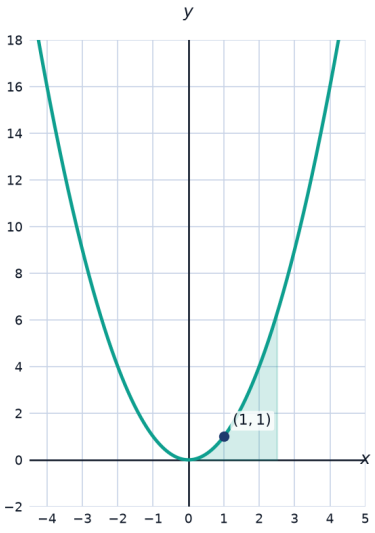
1 [Foundational] • **Target: Area between a curve and the x-axis**

Find the area of the region bounded by $y = 4 - x^2$ and the x-axis.



2 [Foundational] • **Target: Area between two curves (vertical strips)**

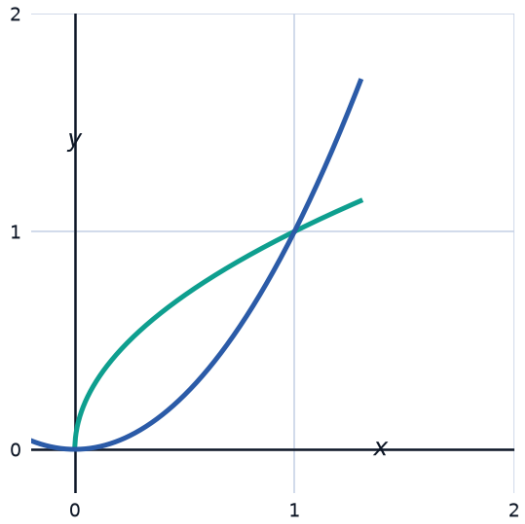
Find the area between $y = x^2$ and $y = x + 2$.



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3 [Foundational] • Target: Area between two curves (vertical strips)

Find the area between $y = \sqrt{x}$ and $y = x^2$ on $[0, 1]$.



4 [Foundational] • Target: Regions requiring horizontal strips

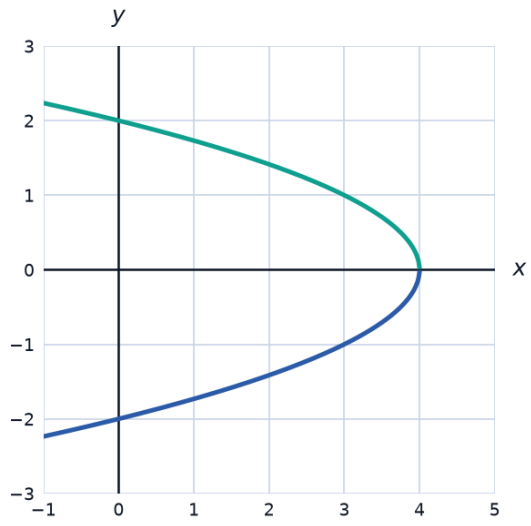
Find the area of the region bounded by $x = y^2$ and $x = y + 2$, integrating with respect to y.

5 [Foundational] • Target: Regions with multiple intersection points

Find the total area between $y = \sin x$ and $y = \cos x$ on $[0, \pi]$, accounting for the curves crossing at $x = \frac{\pi}{4}$.

6 [Foundational] • Target: Area using the more efficient variable of integration

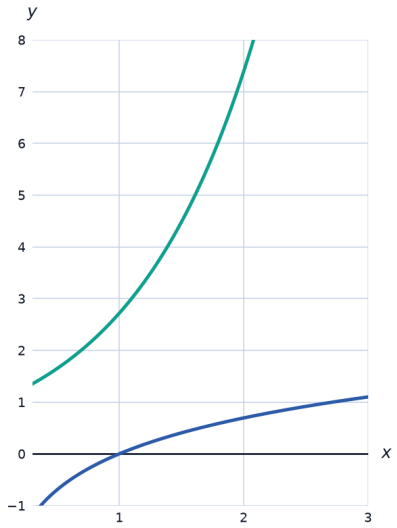
Find the area enclosed by $y^2 = 4 - x$ and $x = 0$, choosing the more convenient variable of integration.



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7 [Foundational] • Target: Setting up but not evaluating (AP-style)

Set up (but do not evaluate) the integral expression for the area of the region enclosed by $y = e^x$, $y = \ln x$, and the lines $x = 1$ and $x = 2$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Setting up but not evaluating (AP-style)

Set up (but do not evaluate) the integral expression for the area between $y = x^3 - 4x$ and the x-axis on $[-2, 2]$, accounting for the curve crossing the axis at $x = -2, 0, 2$.

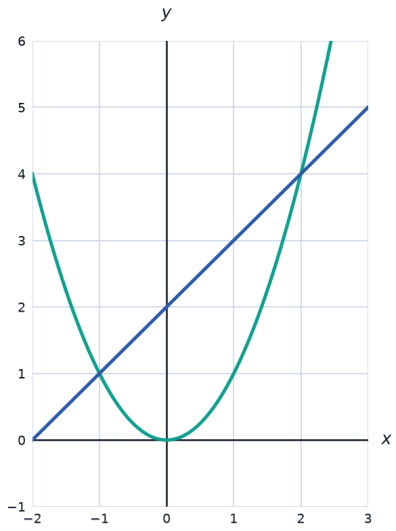
9 [Intermediate] • Target: Area between curves and a table of values

Functions f and g satisfy $f(x) \geq g(x)$ for all x in $[0, 6]$, with selected values given: $f(0) = 10$, $f(2) = 8$, $f(4) = 9$, $f(6) = 12$ and $g(0) = 3$, $g(2) = 4$, $g(4) = 5$, $g(6) = 6$. Use the Trapezoidal Rule with these 3 equal subintervals to approximate the area between the curves, $\int_0^6 [f(x) - g(x)]\,dx$.

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10 [Intermediate] • Target: Visualizing area between two curves

The graph shows $y = x^2$ and $y = x + 2$ from an earlier problem, together with the region between them.



11 [Intermediate] • Target: A capstone area problem

Find the area of the region bounded by $y = x^3 - x$ and $y = 0$ on $[-1, 1]$, accounting for the sign change at $x = 0$.

12 [Intermediate] • Target: Area between an exponential curve and a line

Find the area between $y = e^x$ and $y = x + 1$ on $[0, 2]$ (note $e^x \geq x + 1$ for all x, tangent only at $x = 0$).

13 [Intermediate] • Target: An applied area-between-curves problem

Two competing companies' profit rates (in thousands of dollars per year) are $P_1(t) = 50 - 2t$ and $P_2(t) = 30 + t$ over the first 10 years. Find the total difference in cumulative profit between the two companies over this period, using the area between the curves.

14 [Intermediate] • Target: Regions requiring horizontal strips

Level 2 · Medium Use the method of cylindrical shells to find the volume generated by revolving the region bounded by $y = 3x - x^2$ and $y = 0$ about the y-axis.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

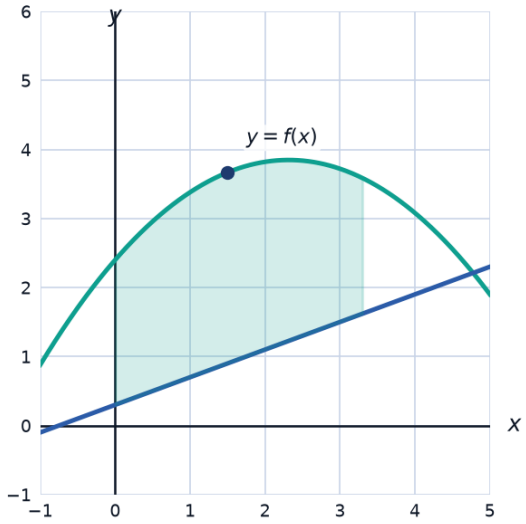
15 [Challenging] • Target: Regions with multiple intersection points • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Level 1 · Foundations Find the area of the region bounded by $y = x^2$ and $y = 2x - x^2$.

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16 **[Challenging] • Target: Area using the more efficient variable of integration • Cross-Lesson: 1.01 Limits Graphically and Numerically**

The region bounded by $y = \ln x$, the line $y = 0$, the line $y = 1$, and the y-axis is rotated 2π radians about the y-axis. Find the exact volume of the resulting solid.



17 **[Challenging] • Target: Setting up but not evaluating (AP-style) • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem**

Two independent random variables are normally distributed as $X \sim N(50, 16)$ and $Y \sim N(45, 9)$. Find the probability that $X > Y$.

18 **[Challenging] • Target: Area between curves and a table of values • Cross-Lesson: 2.01 The Definition of the Derivative**

Level 1 · FoundationsEvaluate the improper integral $\int_1^\infty \frac{1}{x^3}\,dx$ using proper limit notation, or state that it diverges.

19 **[Challenging] • Target: Visualizing area between two curves • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives**

Level 1 · FoundationsEvaluate the indefinite integral $\int \arctan(x)\,dx$.

20 **[Challenging] • Target: A capstone area problem • Cross-Lesson: 3.01 The Chain Rule**

Find the volume of the solid generated when the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is revolved 2π radians about the x-axis.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Area between a curve and the x-axis

Find the total area between $y = x^3 - 36x$ and the x-axis on $[-6, 6]$, accounting for the sign change at $x = -6, 0, 6$.

a. 324

b. 654

c. 108

d. 648

22 [Foundational] • Target: Area between two curves (vertical strips)

Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 3]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 8$, $h(1) = 15$, $h(2) = 6$, $h(3) = 10$. Use the Trapezoidal Rule with these 3 equal subintervals (width 1) to approximate the area between the curves.

a. 39

b. 29

c. 31

d. 30

23 [Foundational] • Target: Regions requiring horizontal strips

Find the area between $y = e^{-2x}$ and $y = -2x + 1$ on $[0, 1]$ (note $e^{-2x} \geq -2x + 1$ for all x, tangent only at $x = 0$), rounded to two decimal places.

a. 0.43

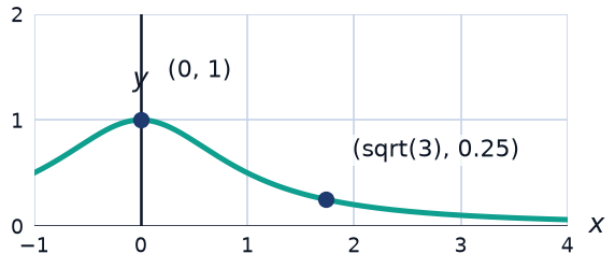
b. 0.52

c. 0.35

d. 1.43

24 [Foundational] • Target: Regions with multiple intersection points

Consider the region bounded by the curve $y = \frac{1}{x^2 + 1}$, the x-axis, and the vertical lines $x = 0$ and $x = b$, where $b > 0$. If the area of this region is $\frac{\pi}{3}$, what is the value of b?



a. $\sqrt{2}$

b. $\frac{1}{\sqrt{3}}$

c. $\sqrt{3}$

d. 2

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25 [Foundational] • Target: Area using the more efficient variable of integration

Consider the region bounded by the curve $y = x^2$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. What is the area of this region?

a. $\frac{26}{3}$

b. $\frac{28}{3}$

c. $\frac{25}{3}$

d. $\frac{27}{3}$

26 [Foundational] • Target: Setting up but not evaluating (AP-style)

Consider the region R in the first quadrant bounded by the curve $y = e^{-x^2}$, the x-axis, and the line $x = a$, where $a > 0$. A solid is generated by revolving this region R about the y-axis. If the volume of this solid is $\frac{\pi}{2}$, determine the value of a.

a. $\ln(2)$

b. $\sqrt{\ln(2)}$

c. $e^{\sqrt{2}}$

d. $\sqrt{2}$

27 [Foundational] • Target: Area between curves and a table of values

Consider the region R in the first quadrant bounded by the curves $y = \sqrt{x}$, $y = 0$, and $x = 4$. A solid is formed by revolving R about the line $x = 5$. Determine the volume of this solid using the method of cylindrical shells. The volume can be expressed in the form $V = k\pi\int_a^b (5 - x)\sqrt{x}\,dx$. Find the value of k.

a. 2

b. 4/3

c. 8/3

d. 16/3

28 [Foundational] • Target: Visualizing area between two curves

$\int_a^b [f(x) - g(x)]dx$ gives area between curves when:

a. $f > g$ and $f < g$ alternates

b. $f > 0$ and $g < 0$

c. $f(x) \geq g(x)$ on $[a, b]$

d. always

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: A capstone area problem**

Find the volume when the area between $y = \sec x$ and $y = \cos x$ (from $-\pi/4$ to $\pi/4$) is rotated about the x-axis:

a. $\pi^2/4$

b. $2\pi(\pi/4 - 1/2)$

c. $\pi(2 - \pi/2)$

d. $\pi(\sqrt{2} - 1)$

30 **[Intermediate]** • **Target: Area between an exponential curve and a line**

Polar area between two curves $r_1(\theta) \le r_2(\theta)$:

a. $\frac{1}{2}\int (r_2^2 - r_1^2)d\theta$

b. $\frac{1}{2}\int (r_2 - r_1)^2 d\theta$

c. $\int (r_2 - r_1)d\theta$

d. $\pi(r_2^2 - r_1^2)$

31 **[Intermediate]** • **Target: An applied area-between-curves problem**

Area between $y = f(x)$ and x-axis on $[a, b]$ equals:

a. $\int_a^b |f(x)|\,dx$

b. $\int_a^b f(x)\,dx$ only when $f \ge 0$

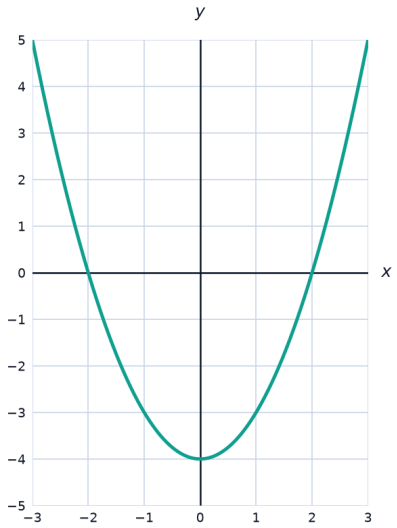
c. $f(b) - f(a)$

d. $\int_a^b f(x)^2\,dx$

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32 **[Intermediate]** • **Target: Area between a curve and the x-axis**

The region bounded by the curve $y = x^2 - 4$ and the x-axis is rotated about the x-axis. If the volume of the resulting solid is $V = \pi\int_a^b (x^2 - 4)^2 dx$, find the value of $b - a$.



a. 2

b. 4

c. 8

d. 16

33 **[Intermediate]** • **Target: Area between two curves (vertical strips)**

Area below x-axis from -2 to 0 of $f(x) = x^2 - 4$:

a. $-\frac{16}{3}$

b. $\frac{8}{3}$

c. $\frac{16}{3}$

d. 4

34 **[Intermediate]** • **Target: Regions requiring horizontal strips**

Tangent line to parametric curve is horizontal when:

a. $dx/dt = 0$

b. $dy/dt = 0$ and $dx/dte0$

c. $t = 0$

d. $dy/dt = dx/dt$

35 **[Intermediate]** • **Target: Regions with multiple intersection points**

Tangent line to parametric curve is vertical when:

a. $dy/dt = 0$

b. $dx/dt = 0$ and $dy/dte0$

c. $dx/dt = dy/dt$

d. $t = 0$

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36 [Intermediate] • Target: Area using the more efficient variable of integration

Area between $y = x$ and $y = x^2$: intersect at $x =$

a. 0 and 1

b. 1 and 2

c. 0 and 2

d. -1 and 1

37 [Intermediate] • Target: Setting up but not evaluating (AP-style)

Signed area and winding number: the winding number of curve C around origin is $\frac{1}{2\pi}\oint_C \frac{x\,dy - y\,dx}{x^2 + y^2} =$

a. always 1

b. an integer

c. always 0

d. always π

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Area between curves and a table of values • Cross-Lesson: 2.01 The Definition of the Derivative

Volume of the solid of intersection of two cylinders $x^2 + z^2 \le 1$ and $y^2 + z^2 \le 1$:

a. $\frac{8}{3}$

b. $\frac{16}{3}$

c. π

d. 4

39 [Challenging] • Target: Visualizing area between two curves • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

The region between $y = x$ and $y = x^3$ in the first quadrant is revolved about the y-axis. Find the volume.

a. $\frac{4\pi}{21}$

b. $\frac{4\pi}{15}$

c. $\frac{2\pi}{15}$

d. $\frac{4}{15}$

40 [Challenging] • Target: A capstone area problem • Cross-Lesson: 3.01 The Chain Rule

Area of triangle with vertices at origin, $(3, 0)$, $(0, 4)$ using integration:

a. 3

b. 12

c. 6

d. π

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41 [Challenging] • Target: Area between an exponential curve and a line • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Total area between $y = x$ and x-axis on $[-2, 2]$:

a. 0

b. 2

c. 4

d. 8

42 [Challenging] • Target: An applied area-between-curves problem • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Moments of inertia: second moment of area $I_x = \int y^2\, dA$ for rectangle $b \times h$ about its centroidal axis:

a. $bh^3/12$

b. $bh^3/3$

c. $bh^2/12$

d. $b^2h/12$

43 [Challenging] • Target: Area between a curve and the x-axis • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Surface area of $y = \sqrt{x}$ revolved about the x-axis from 0 to 4:

a. $\frac{\pi}{3}(17\sqrt{17} - 1)$

b. $\pi(\sqrt{17} - 1)$

c. 8π

d. $\frac{\pi}{6}(17\sqrt{17} - 1)$

44 [Challenging] • Target: Area between two curves (vertical strips) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

For polar curve $r = e^{\theta}$, $\theta \in [0, 2\pi]$, the area swept is:

a. $e^{4\pi}$

b. $\frac{1}{2}(e^{4\pi} - 1)$

c. $\frac{1}{4}(e^{4\pi} - 1)$

d. $\pi e^{4\pi}$

45 [Challenging] • Target: Regions requiring horizontal strips • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Surface area of Gabriel's horn $y = 1/x$, $x \geq 1$, rotated about x-axis:

a. diverges

b. π

c. 2π

d. finite

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